# Wentao Yuan

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## Experience

**Google DeepMind**, Mountain View, CA — 10/2024 – present
*Research Scientist*
Manager: Fei Xia, Jie Tan
Project: Gemini Robotics Embodied Reasoning

**NVIDIA**, Seattle, WA — 02/2021 – 09/2024
*Research Intern*
Mentor: Adithya Murali, Arsalan Mousavian
Project: Scalable Robot Manipulation in Cluttered Environments

**Meta Reality Labs Research**, Redmond, WA — 07/2020 – 11/2020
*Research Intern*
Mentor: Tanner Schmidt, Zhaoyang Lv
Project: Neural Rendering of Dynamic Scenes

**NVIDIA**, Santa Clara, CA — 06/2019 – 12/2019
*Research Intern*
Mentor: Ben Eckart, Kihwan Kim
Project: Learnable 3D Gaussian Mixture Models

## Education

**University of Washington**, Seattle, WA — 2019 – 2024
*Ph.D. in Computer Science*
Advisor: Dieter Fox

**Carnegie Mellon University**, Pittsburgh, PA — 2017 – 2019
*M.S. in Robotics*
Advisor: Martial Hebert

**Pomona College**, Claremont, CA — 2013 – 2017
*B.A. in Computer Science and Mathematics (Summa Cum Laude)*

## Publications

Gemini Robotics Team, **Wentao Yuan** et al. Gemini robotics 1.5: Pushing the frontier of generalist robots with advanced embodied reasoning, thinking, and motion transfer. *arXiv preprint arXiv:2510.03342*, 2025.

Gemini Robotics Team, **Wentao Yuan** et al. Gemini robotics: Bringing ai into the physical world. *arXiv preprint arXiv:2503.20020*, 2025.

**Wentao Yuan**, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithya Murali, Arsalan Mousavian, Dieter Fox. RoboPoint: A Vision-Language Model for Spatial Affordance Prediction for Robotics. In *Conference on Robot Learning (CoRL)*, 2024

**Wentao Yuan**, Adithya Murali, Arsalan Mousavian, Dieter Fox. M2T2: Multi-Task Masked Transformer for Object-centric Pick and Place. In *Conference on Robot Learning (CoRL)*, 2023.

**Wentao Yuan**, Chris Paxton, Karthik Desingh, Dieter Fox. SORNet: Spatial Object-Centric Representations for Sequential Manipulation. In *Conference on Robot Learning (CoRL)*, 2021 (**oral**). **Best Systems Paper Finalist**.

## Publications

**Wentao Yuan**, Zhaoyang Lv, Tanner Schmidt, Steven Lovegrove. STaR: Self-supervised Tracking and Reconstruction of Rigid Objects in Motion with Neural Rendering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Ben Eckart, **Wentao Yuan**, Chao Liu, Jan Kautz. Self-Supervised Learning on 3D Point Clouds by Learning Latent Generative Models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

**Wentao Yuan**, Ben Eckart, Kihwan Kim, Varun Jampani, Dieter Fox, Jan Kautz. DeepGMR: Learning Latent Gaussian Mixture Models for Registration. In *European Conference on Computer Vision (ECCV)*, 2020 (**spotlight**).

**Wentao Yuan**, David Held, Christoph Mertz, Martial Hebert. Iterative Transformer Network for 3D Point Cloud. In *CVPR Workshop on 3D Scene Understanding for Vision, Graphics, and Robotics*, 2019.

**Wentao Yuan**, Tejas Khot, David Held, Christoph Mertz, Martial Hebert. PCN: Point Completion Network. In *International Conference on 3D Vision (3DV)*, Verona, Italy, 2018 (**oral**). **Best Paper Honorable Mention**.

Long Cheng, Jiafei Duan, Yi Ru Wang, Haoquan Fang, Boyang Li, Yushan Huang, Elvis Wang, Ainaz Eftekhar, Jason Lee, **Wentao Yuan**, Rose Hendrix, Noah A. Smith, Fei Xia, Dieter Fox, and Ranjay Krishna. PointArena: Probing Multimodal Grounding Through Language-Guided Pointing. *arXiv preprint arXiv:2505.09990*, 2025.

Adithya Murali, Balakumar Sundaralingam, Yu-Wei Chao, **Wentao Yuan**, Jun Yamada, Mark Carlson, Fabio Ramos, Stan Birchfield, Dieter Fox, and Clemens Eppner. Graspgen: A diffusion-based framework for 6-dof grasping with on-generator training. *arXiv preprint arXiv:2507.13097*, 2025.

Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, **Wentao Yuan**, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in Robotic Manipulation. In *International Conference on Learning Representations (ICLR)*, 2025..

Jiafei Duan, **Wentao Yuan**, Wilbert Pumacay, Yi Ru Wang, Kiana Ehsani, Dieter Fox, and Ranjay Krishna. Manipulate-Anything: Automating Real-World Robots using Vision-Language Models. In *Conference on Robot Learning (CoRL)*, 2024.

Adam Fishman, Aaron Walsman, Mohak Bhardwaj, **Wentao Yuan**, Balakumar Sundaralingam, Byron Boots, and Dieter Fox. Avoid Everything: Model-Free Collision Avoidance with Expert-Guided Fine-Tuning. In *Conference on Robot Learning (CoRL)*, 2024.

Chahyon Ku, Carl Winge, Ryan Diaz, **Wentao Yuan**, Karthik Desingh. Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning. In *International Conference on Robotics and Automation (ICRA)*, 2024.

Xiangyun Meng, Nathan Hatch, Alexander Lambert, Anqi Li, Nolan Wagener, Matthew Schmittle, JoonHo Lee, **Wentao Yuan**, Zoey Chen, Samuel Deng, Greg Okopal, Dieter Fox, Byron Boots, Amirreza Shaban. TerrainNet: Visual Modeling of Complex Terrain for High-speed, Off-road Navigation. In *Robotics: Science and Systems (RSS)*, 2023.

Jeffery Rutledge*, **Wentao Yuan***, Jane Wu, Zoe Wood, Timmy Gambin, Christopher Clark. Intelligent Shipwreck Search Using Autonomous Underwater Vehicles. In *International Conference on Robotics and Automation (ICRA)*, 2018.